NOTRE DAME UNIVERSITY

BANGLADESH

EEE Lab Report-03

Course Title: Electrical Circuit Analysis Lab

Course Code: EEE-2108

Experiment No: 03

Name of the Experiment: Design of Clippers

Date of Experiment: 10-08-23

Date of Submission: 17-08-23

Submitted by:

Group: E

Name: Istiak Alam

ID: 0692230005101005

Batch: CSE-20

Submitted To:

Dhiman Roy

Lecturer, NDUB

Experiment No 3

Name of experiment: Design of Clippers.

Theory: Clippers remove signal voltage above & below a specified level. In the experiment no 2, half wave rectifier can also be called as clipper circuit. Because it clipped off the negative half cycle of the input signal.

A diode connected in series with the load can clipped off any half cycle of input depending on the orientation of the diode.

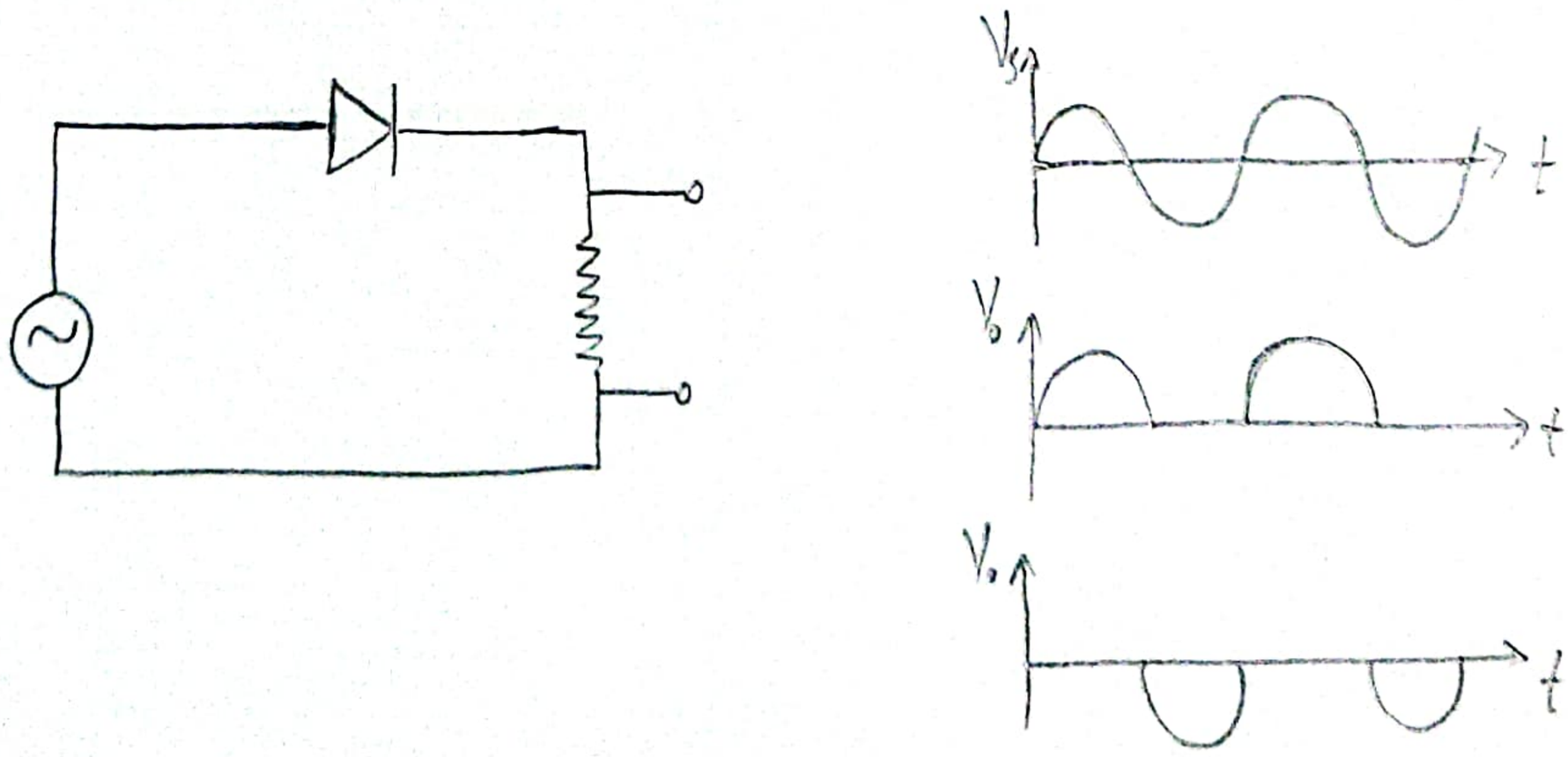

Fig 3.1 : Simple Diode Clippers

A diode connected in series.

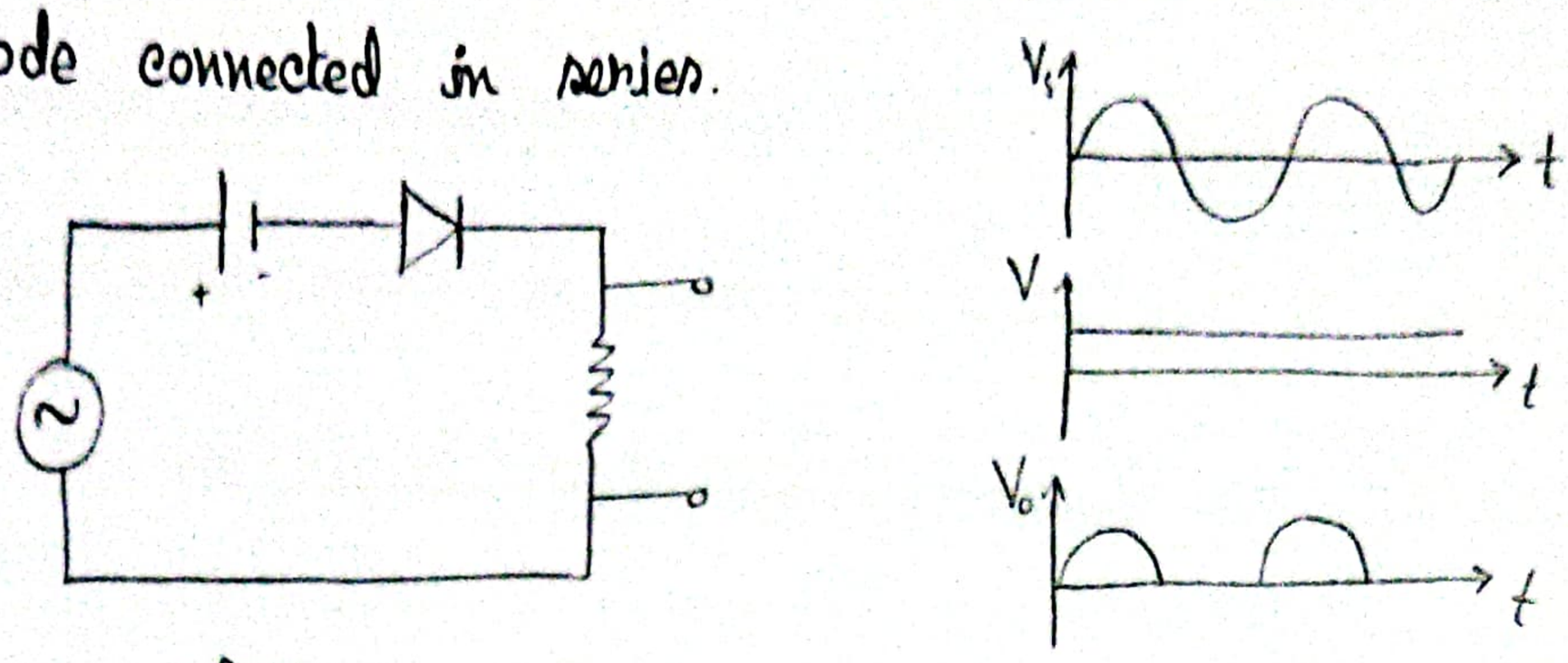

Fig 3.2: Clipper circuit using Bias Diode.

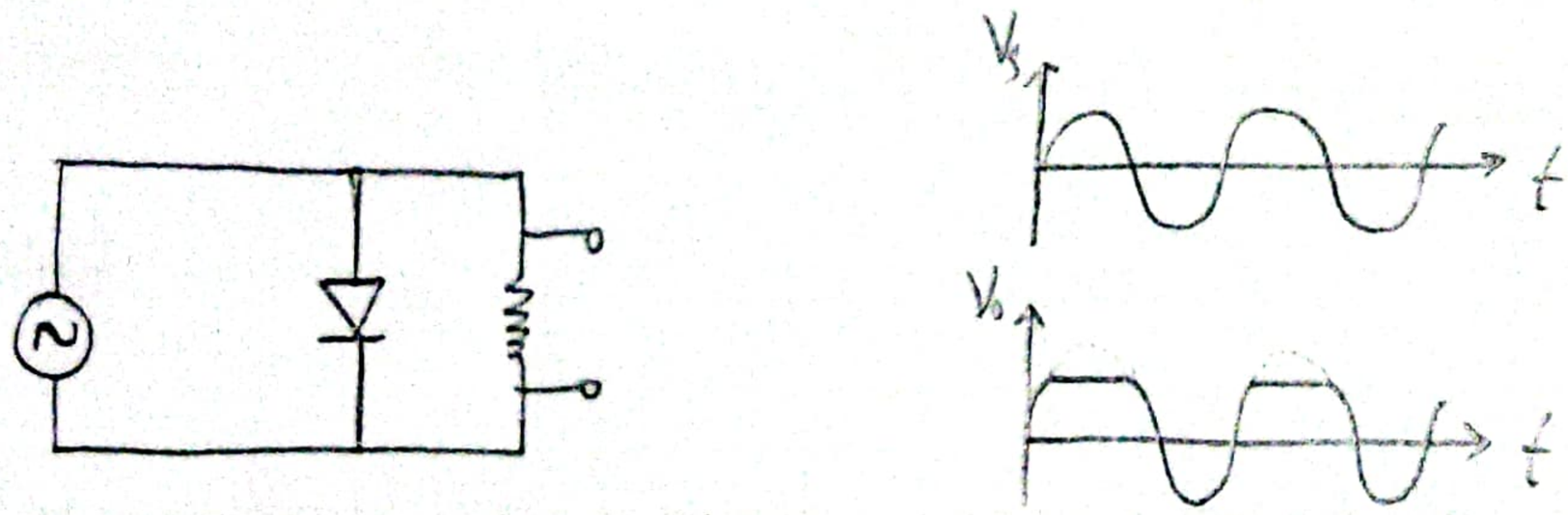

Fig 3.3: Parrallel Clipper Circuit.

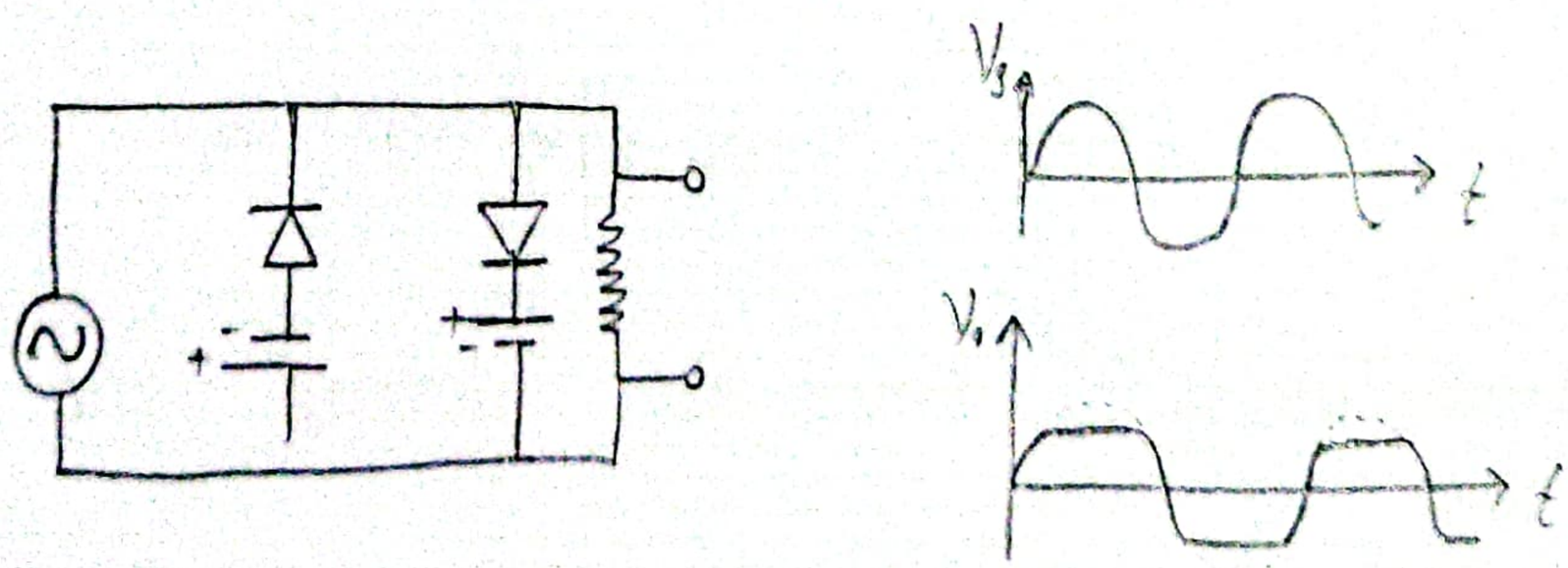

Fig 3.4: Biased Parallel Clipper Circuit.

Procedures:

(i) Determine the direction of the diode. It depends on the direction of the half cycle in which the peak can be seen or the peek is not clipped. Constracting the circuit like fig 3.5, 3.6, 3.7.

(ii) If any half-cycle is clipped or has a certain DC value, then the polarity of the biasing DC voltage source can be determined.

(iii) After constract the circuit we ~~conect~~ connect the channel one to the two point of resistance & ~~two~~ connect channel two to the input point of signal generator.

(iv) The signal generator will provide AC voltage through the circuit & we will found the input & output curve in the oscilloscope.

Clipper Circuit & Graph:

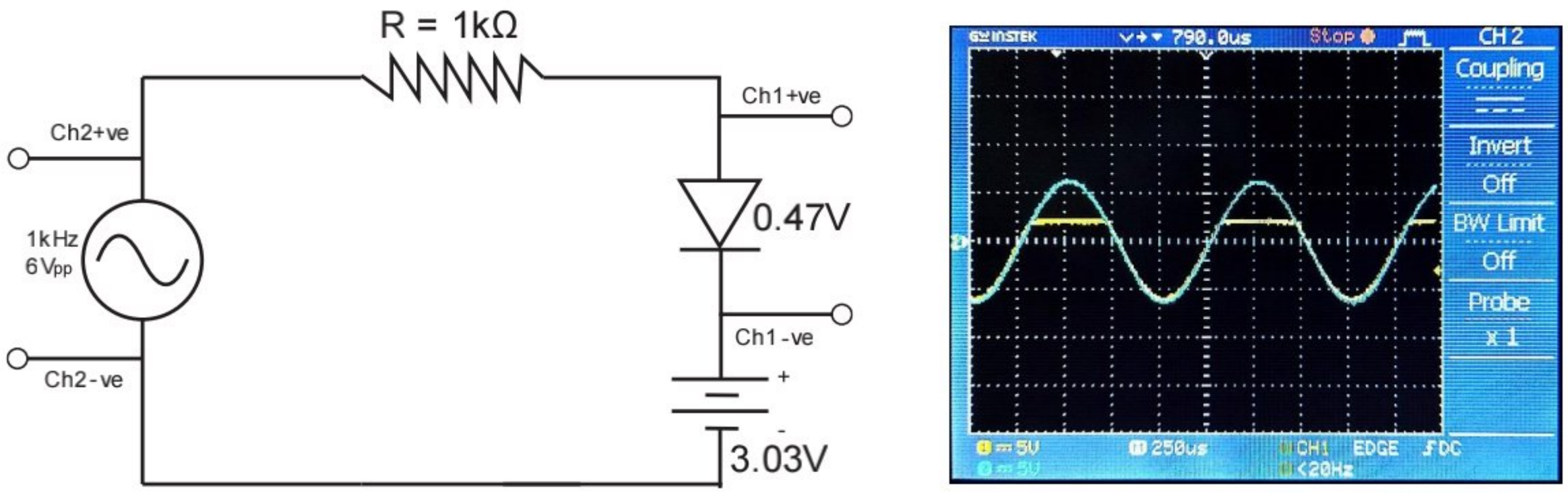

Fig 3.5: Biased Clipper Circuit & Wave Graph (Positive half cycle is clipped)

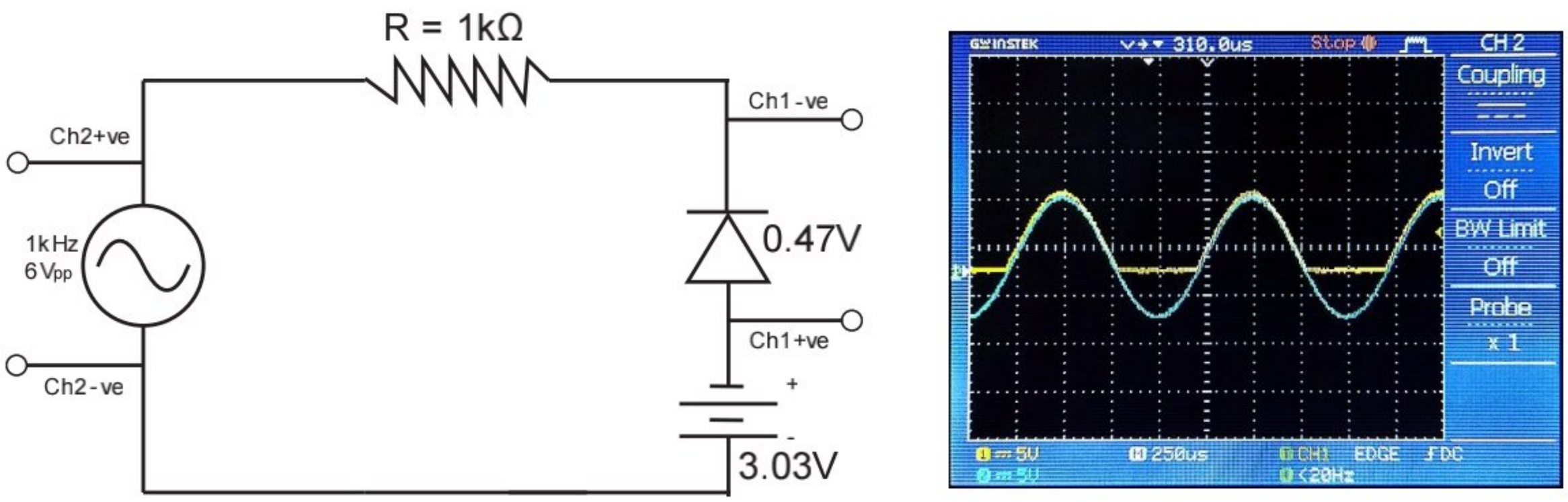

Fig 3.6: Biased Clipper Circuit & Wave Graph (Negative half cycle is clipped)

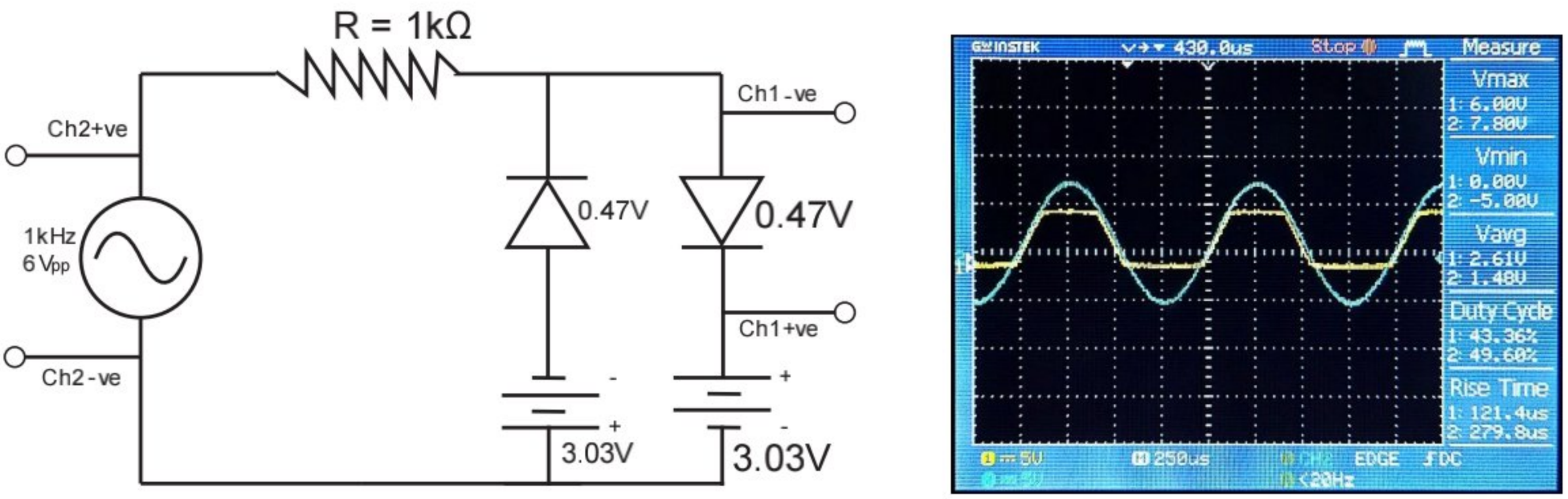

Fig 3.7: Biased Parallel Clipper Circuit & Wave Graph

Discussion:

(i) When supplying the AC Signal. It's required a steady DC Current or voltage to operate correctly.

(ii) Avoiding overload & short circuit

(iii) Follow guidlines & manual